

Diet-Related Parasitic Infections in Cuban Tree Frogs: Insights from Stomach and Intestinal Content

Jose Ledea, Dr. Christina Anaya

Biological Sciences, Florida Gulf Coast University

ABSTRACT

The Cuban tree frog (*Osteopilus septentrionalis*) is an invasive species in Florida, known for its significant ecological impacts, including disruption of native ecosystems, competition with native amphibians and reptiles, and the potential to introduce novel parasites. Despite extensive documentation of their diet, the link between dietary diversity and parasitic infections in this species remains poorly understood. This study investigates the relationship between prey consumption and parasite prevalence in Cuban tree frogs collected from Southwest Florida.

Thirty frogs were dissected, and their stomach and intestinal contents were examined to identify prey and parasite loads. Female frogs, which consumed larger and more diverse prey than males, exhibited significantly higher parasite variability. The most common prey items included beetles, caterpillars, and spiders, with larger prey items correlating to increased parasitic infections. This study underscores the ecological threat posed by Cuban tree frogs, particularly their role as a vector for parasites and their competition with native species for prey. Future management strategies should prioritize targeted removal efforts and community awareness campaigns to mitigate the ecological consequences of this invasive species.

INTRODUCTION

Invasive species significantly disrupt ecosystems worldwide by altering food webs, outcompeting native organisms, and introducing pathogens (Simberloff & Rejmánek, 2011; Vitousek et al., 1997). The Cuban tree frog (*Osteopilus septentrionalis*), native to the Caribbean, exemplifies these disruptions. First documented in Florida in the 1920s, this species has established itself as a formidable competitor to native amphibians and reptiles due to its generalist feeding habits and ability to thrive in diverse habitats, including wetlands and urban areas (Dameron, 2016; Johnson et al., 2023).

Cuban tree frogs are opportunistic predators, preying on a wide range of invertebrates, including insects and arachnids, as well as smaller vertebrates like native frogs and lizards (Wyatt & Forsy, 2004; Meshaka, 1996). This feeding behavior not only depletes resources for native insectivores but also exposes Cuban tree frogs to parasites that can spill over into native populations, further disrupting ecosystem dynamics (Lykins et al., 2024; Poulin, 2007).

Parasites, such as nematodes and trematodes, are known to influence host behavior, reproduction, and survival, amplifying the ecological consequences of invasive species (Sutherland, 2005).

While previous studies have documented the diet and ecological impacts of Cuban tree frogs, there is limited research on the link between their dietary habits and parasite loads. Factors such as prey diversity and size may influence parasite acquisition, particularly in female frogs, which require higher energy intake for reproduction (Knutie et al., 2017). Additionally, the broader ecological risks posed by Cuban tree frogs, including their potential to alter nutrient

cycling and act as disease vectors, are significant concerns (U.S. Fish and Wildlife Service, 2014; Melissa, 2017).

This study investigates the dietary habits and parasite prevalence in male and female Cuban tree frogs from Southwest Florida. By analyzing stomach and intestinal contents, we aim to understand how prey diversity and consumption relate to parasitic infections, providing insights for management strategies to mitigate the ecological impacts of this invasive species.

MATERIALS AND METHODS

Thirty Cuban tree frogs were collected by independent contractors from three sites in Southwest Florida. Site A (26.46541° N, 81.77423° W): 11 frogs, site B (26.57354° N, 81.82378° W): 16 frogs, and site C (26.34672° N, 81.81177° W): 3 frogs found. Frogs were euthanized humanely and stored at -20°C until dissection.

Thawed frogs were weighed using an OHAUS Compass scale and measured for snout-to-vent length (SVL) with a flexible measuring tape. Morphological characteristics, such as vocal sacs and body size, determined sex, which was later confirmed during dissection.

Stomach, intestinal, and colon contents were removed using surgical scalpels and scissors and placed in glass petri dishes filled with water for examination. Stomach contents were decanted to separate undigested debris, while intestinal tracts were examined for parasites. A stereo dissecting microscope (Olympus SZX7) with varying light settings was used to identify prey items, which were then preserved in a falcon tube filled with 70% ethanol. Identification was performed using resources such as BugGuide.net and the *Field Guide to Diseases and Insects of Quaking Aspen in the West*.

Prey items were categorized by type (e.g., beetles, caterpillars, spiders) and size. Parasite counts were recorded, and variability in loads between sexes was analyzed using T-tests and correlation analyses. Excel was used for statistical calculations and data visualization to explore potential relationships between dietary habits and parasite load. Macroparasites were sought across the digestive section of the stomach; however, none were observed during the examinations. These methods aimed to provide insights into the Cuban tree frogs' diet, prey diversity, and parasitism patterns.

RESULTS

A total of 30 Cuban tree frogs (*Osteopilus septentrionalis*) were examined, including 9 males and 21 females. The average mass for males was $4.23 \text{ g} \pm 1.10$ (range: 2.2–5.4 g), and their average snout-to-vent length was $4.66 \text{ cm} \pm 0.46$ (range: 4–5.5 cm). Female frogs had a significantly higher average mass of $14.0 \text{ g} \pm 7.69$ (range: 3.2–31.9 g) and length of $6.50 \text{ cm} \pm 1.11$ (range: 5–8.5 cm) ($t = 5.67$, $p < 0.01$ for mass; $t = 4.35$, $p < 0.01$ for length).

Prey consumption also differed significantly between sexes. Females consumed more prey items than males, with a mean of 3.93 ± 0.76 (range: 1–11) prey items per female compared to 1.36 ± 0.52 (range: 1–6) prey items per male ($t = 3.48$, $p < 0.01$). Prey items were identified and categorized, with beetles, caterpillars, and bugs being the most common types. Females demonstrated a preference for leaf and ground beetles, while males primarily consumed true bugs and beetles, with fewer overall prey types.

Parasite prevalence was higher in females. The average parasite load in females was 26.86 ± 15.24 (range: 0–200), which was higher than the average parasite load in males (16.7 ± 9.93 ; range: 0–118). However, this difference was not statistically significant ($t = 1.15$, $p = 0.26$).

Leaf beetles and true bugs were most strongly associated with higher parasite counts. Females exhibited greater variability in parasite loads, likely reflecting their more diverse diets. Parasites were predominantly found in the intestines and colon, with larger prey items, such as beetles and grasshoppers, also correlating with increased parasite counts. Correlation analyses showed no significant relationship between frog length and either prey consumption ($R^2 = 0.0108$) or parasite load ($R^2 = 0.0089$). These findings suggest that while sex influences prey consumption and parasite burden, frog size does not.

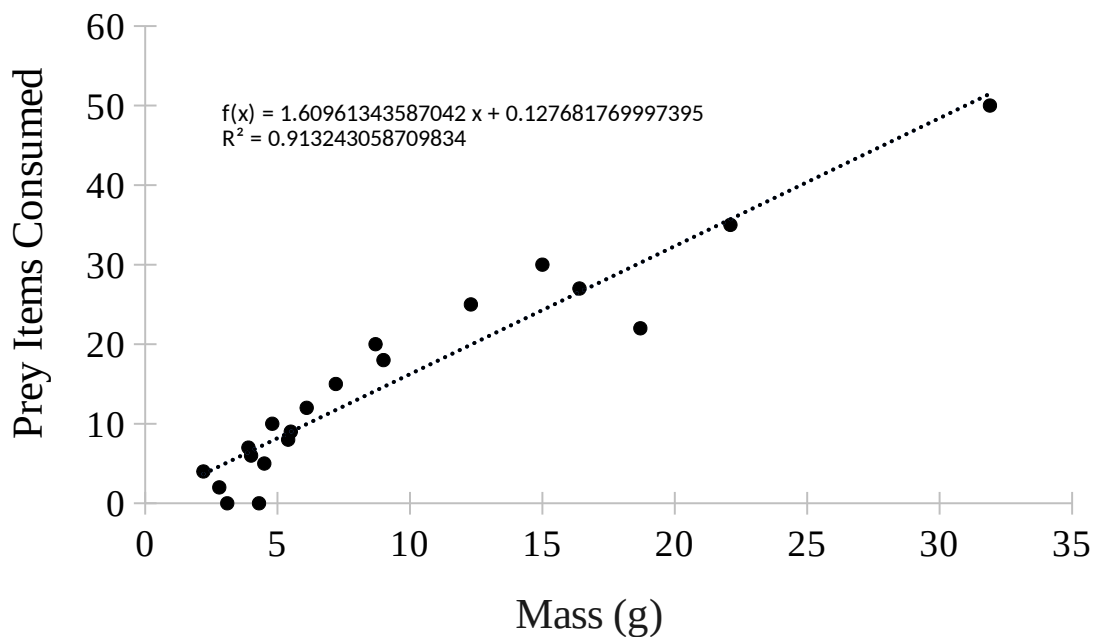


Figure 1: Scatter plot showing correlation between mass and prey items consumed for male and female frogs altogether.

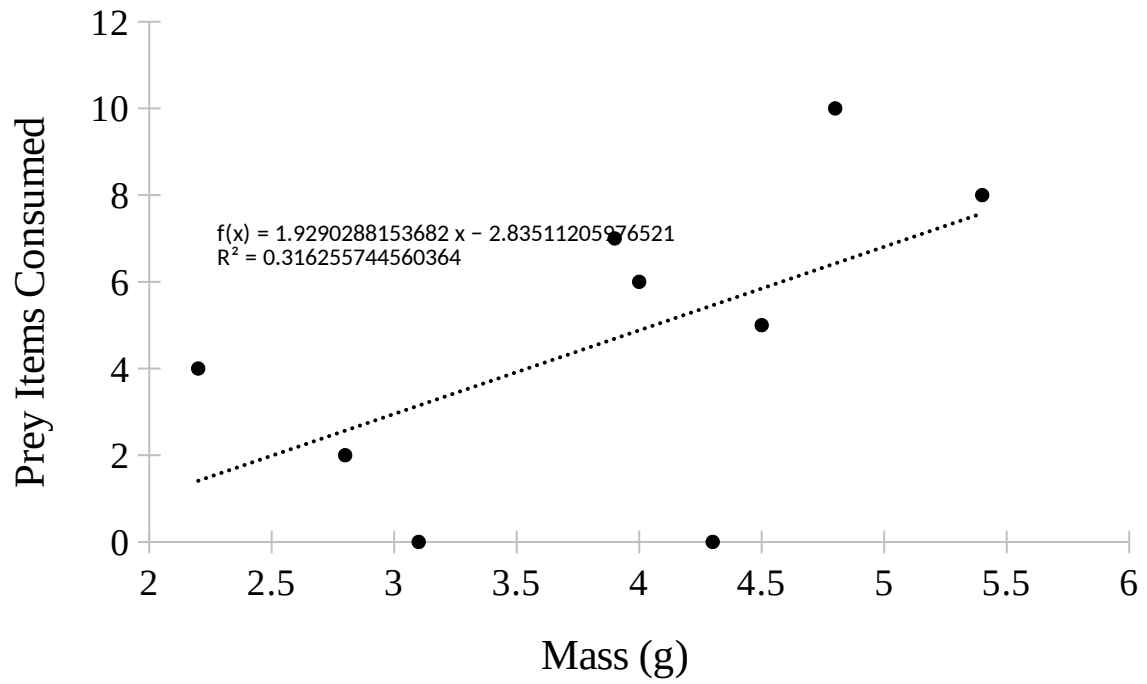


Figure 2: Scatter plot showing correlation between mass and prey items for males only.

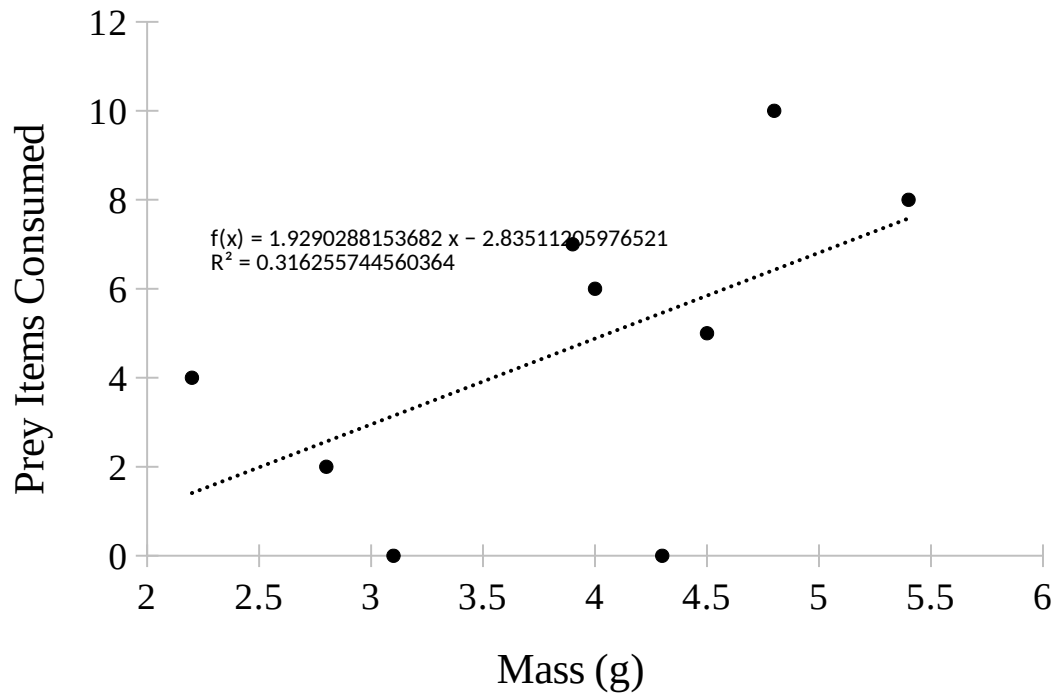


Figure 3: Scatter plot showing correlation between mass and prey items for females only.

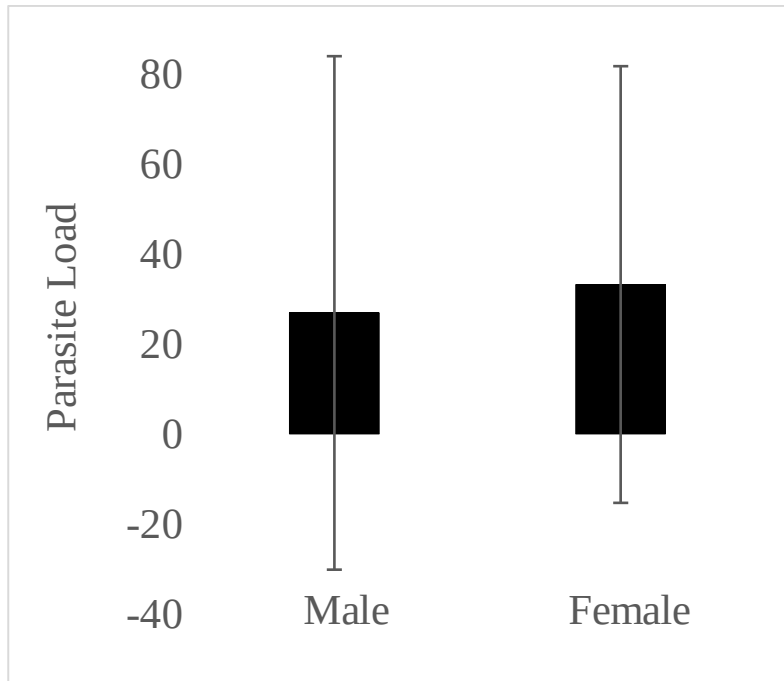


Fig. 4: Comparison of parasite load between male and female Cuban tree frogs. Error bars indicate standard deviation.

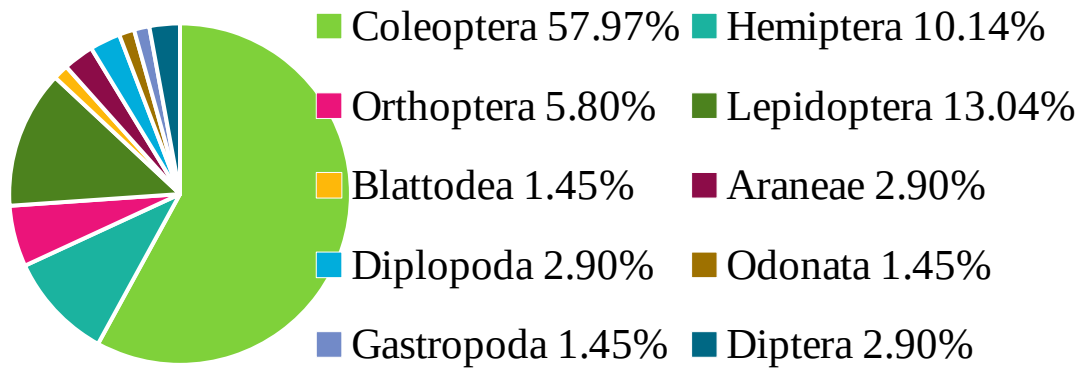


Fig. 5: Proportional representation of prey items in diet of *O. septentrionalis*.

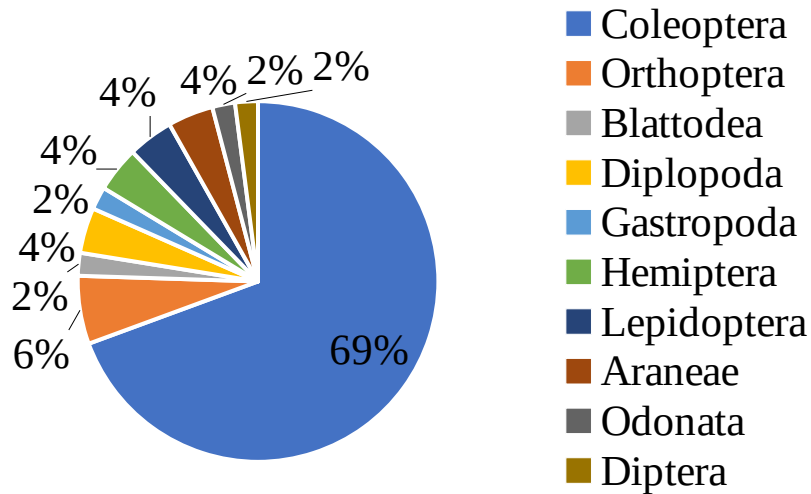


Fig. 6: Dietary composition of female frogs.

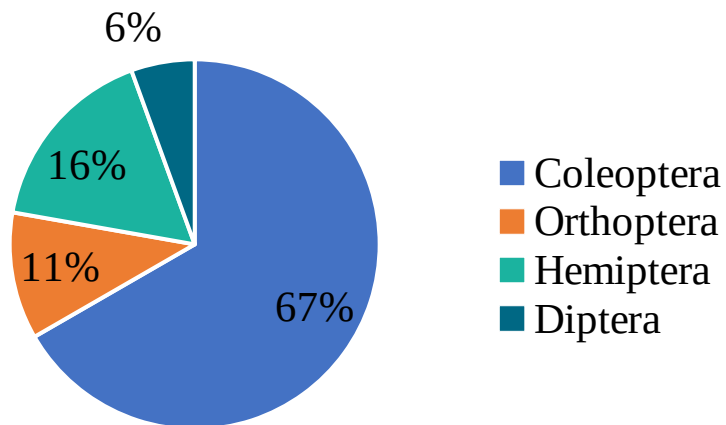


Fig. 7: Dietary composition of male frogs.

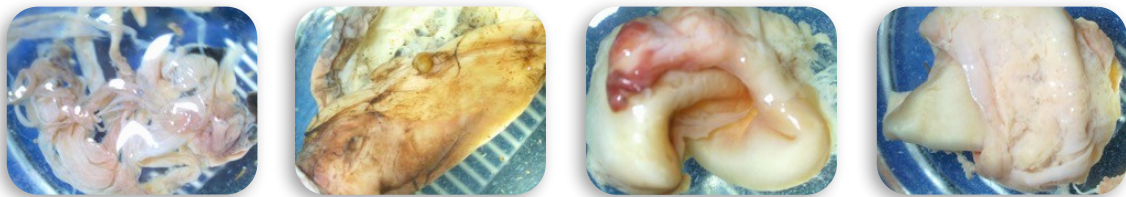


Fig. 8: Internal view of anatomy and composition of *O. septentrionalis* stomachs as observed during dissection.

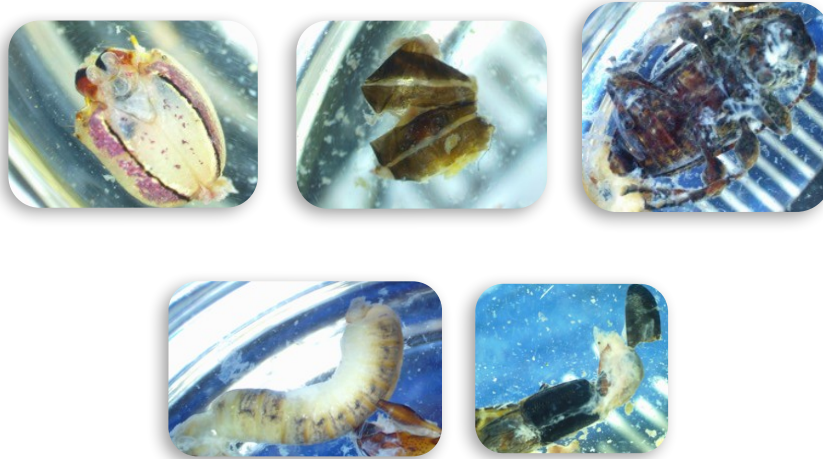


Fig. 9: Coleoptera (Beetles): *Chrysomelidae*, *Cassidinae*, *Phyllophaga* spp. larva, and *Scarabaeidae*.



Fig. 10: Hemiptera (True bugs): *Heteroptera* and *Cicadellidae*.

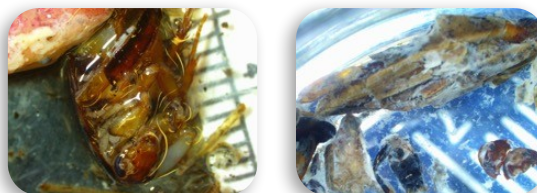


Fig. 11: Orthoptera (Grasshoppers/crickets): *Acrididae* and *Gryllidae*.

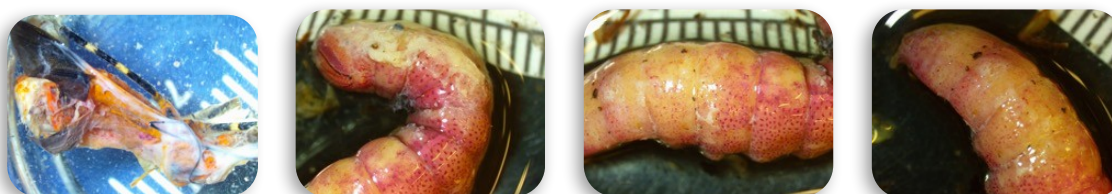


Fig. 11: Lepidoptera (Butterflies, caterpillars): *Lepidoptera* and *Lepidoptera* larvae (head – left, midsection – center, tail – right).



Fig. 12: *Araneae*, *Diplopoda*, *Syrphidae* larvae, and *Gastropoda*.

DISCUSSION

The results of this study revealed significant differences in dietary habits and parasite loads between male and female Cuban tree frogs (*Osteopilus septentrionalis*) collected from Southwest Florida. Female frogs, which exhibited larger body sizes, consumed a greater quantity and diversity of prey items and harbored higher parasite loads compared to males. This correlation suggests that females' larger dietary intake, particularly of arthropods such as beetles and grasshoppers, may contribute to increased susceptibility to parasitic infections. The observed variability in parasite loads among female Cuban tree frogs may be influenced by dietary factors, as studies have shown that changes in diet can affect body size and development in nematodes (So et al., 2012). However, the smaller sample size of males (n=9) compared to females (n=21) limits the ability to draw definitive conclusions about the ratio of parasites between the sexes. This sample imbalance underscores the need for caution in interpreting the observed trends, and further research is needed to determine whether specific prey items are intermediate hosts for parasites or whether habitat-specific prey abundance contributes to these differences. The relationship between prey intake and parasite load could also be influenced by regional microhabitats, which may provide favorable conditions for larger arthropods and parasite proliferation.

The diet of Cuban tree frogs in this study included a wide range of invertebrates, such as beetles, caterpillars, bees, millipedes, roaches, spiders, dragonflies, and grasshoppers. This aligns with prior research highlighting their opportunistic feeding behavior but contrasts with earlier findings that reported a preference for smaller prey items (Meshaka et al., 2000). The prevalence of larger prey items among females in our study could be attributed to their greater energy demands for reproduction (Knutie et al., 2017) or to regional variations in prey availability in Southwest Florida. Notably, the timing of prey consumption and digestion stages remains unknown, complicating interpretations of prey item counts and their direct correlation with parasite loads. Digestion and sample timing may introduce variability, warranting further investigation.

These findings carry significant implications for native species and ecosystem dynamics and align with patterns observed in other biological invasions, where introduced species exhibit complex parasite infection dynamics (Melissa, 2017). Cuban tree frogs, as invasive predators, compete with native insectivores such as frogs, lizards, and small reptiles for invertebrate prey, potentially leading to resource depletion and displacement of native species (Beard et al., 2017). Studies have documented dietary overlap between Cuban tree frogs and native species, further emphasizing the ecological threat posed by their presence (Glorioso et al., 2010). The reduction of invertebrate populations by Cuban tree frogs could also disrupt local food webs, indirectly affecting insectivorous birds and small mammals (Wyatt & Forsys, 2004). These cascading effects highlight the broader ecological impact of this invasive species.

Management strategies targeting Cuban tree frog populations could benefit from insights into their dietary habits and prey preferences. For example, removal efforts could prioritize habitats where Cuban tree frogs consume the largest quantities of prey or exhibit the highest

parasite loads, reducing their population density in areas most affected by their presence. Additionally, public education campaigns could raise awareness of the ecological impact of Cuban tree frogs and encourage community participation in removal efforts (Olšanská et al., 2016).

Future management strategies should prioritize removal efforts in habitats with high prey abundance, where Cuban tree frogs are likely to exhibit greater parasite loads. Efforts should aim to identify the specific prey items that serve as vectors for parasites and assess the long-term impact of Cuban tree frogs on native species and biodiversity. Public education campaigns could also play a critical role in mitigating the spread of this invasive species by raising awareness of their ecological impacts. Understanding the association between prey size, dietary intake, and parasite transmission can inform conservation efforts and strategies to mitigate the ecological effects of this invasive species.

ACKNOWLEDGEMENTS

I express my gratitude to my professor for her invaluable guidance and support throughout this project and to my colleagues for their constructive feedback during the research process. I also thank the independent contractors who assisted with specimen collection. This study was made possible by the resources provided by Florida Gulf Coast University (FGCU), which supported both the laboratory and fieldwork aspects of this research. This project was conducted under FGCU's institutional approval, IACUC# RSC-0097.

LITERATURE CITED

- Babbitt, K. J., & Meshaka, W. E. (2000). Benefits of Eating Conspecifics: Effects of Background Diet on Survival and Metamorphosis in the Cuban Treefrog (*Osteopilus septentrionalis*). *Copeia*, 2000(2), 469–474. <http://www.jstor.org/stable/1448194>
- Beard, K.H., Johnson, S.A., & Shiels, A.B. (2017). Cuban Tree Frogs (*Osteopilus septentrionalis*): Distribution, Diet, and Reproduction of an Invasive Species in Ecology and Management of Terrestrial Vertebrate Invasive Species in the United States. *CRC Press*. DOI: [10.1201/9781315157078-9](https://doi.org/10.1201/9781315157078-9)
- Dameron, M. (2016). *Osteopilus septentrionalis*. *Animal Diversity Web*, https://animaldiversity.org/accounts/Osteopilus_septentrionalis/
- Glorioso, B. M., Waddle, J. H., Crockett, M. E., Rice, K. G., & Percival, H. F. (2010). Diet of the invasive Cuban Treefrog (*Osteopilus septentrionalis*) in pine rockland and mangrove habitats in South Florida. *Caribbean Journal of Science*, 46(2–3), 346-355. DOI: <https://doi.org/10.18475/cjos.v46i2.a25>
- Johnson, S. A., McGarrity, M. E., & Roanoke, E.A. (2023). The Cuban treefrog (*Osteopilus septentrionalis*) in Florida. University of Florida IFAS Extension. Critical Issue: 3. *EDIS*, vol. 2023, no. 1, DOI: <https://doi.org/10.32473/edis-UW259-2023>
- Knutie, S. A., Rohr, J. R., & Kohl, K. D. (2017). Early-Life Diet Affects Host Microbiota and Later-Life Defenses Against Parasites in Frogs. *Integrative and Comparative Biology*, 57(4), 732-742, DOI: <https://doi.org/10.1093/icb/icx028>
- Lykins, K., Ossiboff, R. J., Chase, E., Thompson, N., Farrell, T. M., Wu, T., Johnson, S. A., & Walden, H. D. S. (2024). Pterygodermatites (Mesopectines) whartoni (Nematoda: Rictulariidae) encysted larvae in invasive Cuban treefrogs (*Osteopilus septentrionalis*)

- from Florida, United States. *Frontiers in Veterinary Science*, 11, 1353975. DOI: <https://doi.org/10.3389/fvets.2024.1353975>
- McDonough, K. (2012). Review of the book Encyclopedia of biological invasions, by D. Simberloff & M. Rejmanek. *American Reference Books Annual*, 1175. https://commons.nmu.edu/cgi/viewcontent.cgi?article=1012&context=library_publication
- Melissa, M. A. (2017). Patterns of parasite infection dynamics during a biological invasion. *ProQuest Dissertations and Theses*, 113. <https://www.proquest.com/dissertations-theses/patterns-parasite-infection-dynamics-during/docview/2779137913/se-2>
- Meshaka, W. E. (1996). Diet and the colonization of buildings by the Cuban Treefrog, *Osteopilus septentrionalis* (Anura: Hylidae). *Caribbean Journal of Science*, 32, 59-63. https://scholar.google.com/citations?view_op=view_citation&hl=en&user=HLkw17QAAAAJ&citation_for_view=HLkw17QAAAAJ:qjMakFHDy7sC
- Olšanská, B., Kasarda, R., Trakovická, A., Kadlečík, O., Moravčíková, N. (2020): Identification of selection signals through haplotype structure in genome of dual-purpose breeds. *Agriculture and Forestry*, 66 (1): 33-39. DOI: 10.17707/AgricultForest.66.1.04
- Ortega, N., Price, W., Campbell, T., & Rohr, J. (2015). Acquired and introduced macroparasites of the invasive Cuban treefrog, *Osteopilus septentrionalis*. *International Journal for Parasitology: Parasites and Wildlife*, 4(3), 379-384, DOI: <https://doi.org/10.1016/j.ijppaw.2015.10.002>
- Poulin, R. (2007). Evolutionary Ecology of Parasites: (Second Edition). *Princeton University Press*. <http://www.jstor.org/stable/j.ctt7sn0x>

- Ricciardi, A., Hoopes, M.F., Marchetti, M.P. and Lockwood, J.L. (2013), Progress toward understanding the ecological impacts of nonnative species. *Ecological Monographs*, 83(3): 263-282. <https://doi.org/10.1890/13-0183.1>
- Simberloff, D. & Rejmanek M. (2011). Encyclopedia of Biological Invasions. *University of California Press*, ISBN 13: 978-0-520-26421-2.
<https://www.ucpress.edu/books/encyclopedia-of-biological-invasions/hardcover>
- So, S., Garan, Y., Miyahara, K., & Ohshima, Y. (2012). Body size change in various nematodes depending on bacterial food, sex and growth temperature. *Worm*, 1(2), 93–97.
<https://doi.org/10.4161/worm.20175>
- Sutherland, D. (2005). Parasites of North American frogs. In *Amphibian Declines*. *University of California Press*, DOI: <https://doi.org/10.1525/9780520929432-023>
- U.S. Fish and Wildlife Service. (2014). Cuban Treefrog (*Osteopilus septentrionalis*) Ecological Risk Screening Summary. FWS,
<https://www.fws.gov/sites/default/files/documents/Ecological-Risk-Screening-Summary-Cuban-Treefrog.pdf>
- Vitousek, P. M., Mooney, H. A., Lubchenco J, Melilo J.M. (1997). Human Domination of Earth's Ecosystems. *Science*, vol. 277, issue 5325, pp. 494-499. DOI: 10.1126/science.277.5325.494
- Wyatt, J. L., & Forsy, E. A. (2004). Conservation Implications of Predation by Cuban Treefrogs (*Osteopilus septentrionalis*) on Native Hylids in Florida. *Southeastern Naturalist*, 3(4), 695-700. DOI: [https://doi.org/10.1656/1528-7092\(2004\)003\[0695:CIOPBC\]2.0.CO;2](https://doi.org/10.1656/1528-7092(2004)003[0695:CIOPBC]2.0.CO;2)